Follow-up Analyses of Motor Subtype Progression

Additional work carried out after the research review meeting

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Abstract

After reviewing our earlier work on TD/Indeterminate/PIGD motor subtype progression in the PPMI cohort, several questions were raised about how we were analysing and presenting the results. This report covers only the additional analyses carried out in response, and assumes the earlier work is already familiar.

Five things were asked for: a look at where inside the Indeterminate range those observations actually fall; subtype progression shown for individual patients rather than as a population summary; the conversion analysis repeated in the direction of TD as well as PIGD; the raw measured tremor/PIGD ratio shown directly rather than only through a fitted model; and the separate population-level model plots combined into one.

The Indeterminate ratios were found to be concentrated near the two boundaries of the Indeterminate range: about 78% of the visits with a computable ratio occur at just two values, one slightly above the PIGD threshold and one slightly below the TD threshold. Correcting the patient-level progression analysis to use only the 934 patients who have at least two labelled visits, 476 (51.0%) show no recorded subtype change, 210 (22.5%) change once and 248 (26.6%) change more than once. Conversion is recorded in both directions, and requiring the new subtype to persist to the next visit roughly halves the estimated rate in both cases. The raw ratio shows a downward trend over follow-up, and also moves by more than the entire width of the Indeterminate range between consecutive visits of the same patient in about three quarters of cases.

The Indeterminate distribution is presented here as an observation, for discussion, without further analysis of what follows from it. The remaining results suggest that individual recorded subtype changes should be read cautiously, and that a sustained change carries more weight than a single one.

1 Purpose of the follow-up analysis

The discussion of our earlier results raised a number of questions that we could not answer from the analyses we had already run. They fell into five groups.

The first concerned the Indeterminate category. We had reported that it is unstable, with only about a fifth of Indeterminate visits followed by another Indeterminate visit. The question raised was where inside the Indeterminate range those observations actually sit, and in particular whether they lie close to one of the two subtype boundaries.

The second concerned how we were presenting progression. Our figures summarised the whole population at each follow-up year, which mixes genuine change with the fact that different patients are present at different times. What was wanted was a view of what happens to individual patients.

The third was that our conversion analysis only looked in one direction. We had estimated how long it takes to be recorded as PIGD, but not how often patients move the other way.

The fourth concerned the ratio itself. The subtype label is produced by cutting a continuous quantity into three categories, and our figure of ratio progression showed a fitted model rather than the measurements. The request was to see the raw values.

The fifth was a presentation issue: the model-based population results were spread across three separate panels, which makes the three starting subtypes difficult to compare.

We addressed each of these. None of the analyses below changes the dataset, the subtype definition, the medication filter or any fitted model; they are new views of data we already had, with one correction to a calculation that is described in Section 3.

For reference, the classification rule being used throughout is unchanged:

$$R = \frac{\text{mean of the 11 tremor items}}{\text{mean of the 5 gait and postural items}}, \qquad \begin{cases} R \geq 1.15 & \text{TD} \\ 0.90 < R < 1.15 & \text{Indeterminate} \\ R \leq 0.90 & \text{PIGD} \end{cases}$$

In words: the ratio compares how severe a patient's tremor symptoms are against how severe their gait and balance symptoms are, and two fixed cut-offs turn that comparison into a label.

2 Where inside the Indeterminate range do these observations sit?

The suggestion was to take the visits recorded as Indeterminate, particularly those arriving from TD or from PIGD, and look at their actual ratio values rather than only at how many there are. This section reports what the distribution looks like. We have deliberately not taken the analysis further than that, so that the observation can be discussed before anything is concluded from it.

There are 441 Indeterminate visits in the panel. Of these, 381 have a ratio that can be computed; the remaining 60 have both subscores equal to zero, so the ratio is undefined even though the classification rule still assigns them to Indeterminate. Of the 381, 287 follow an earlier labelled visit and 94 are the patient's first labelled visit, so they have no preceding subtype. Across the whole panel, 332 transitions end in Indeterminate.

The observed ratios are concentrated near the two boundaries of the Indeterminate range. Across all 381 visits only 11 distinct ratio values occur, and two of them account for most of the observations:

- **52.0%** of these visits occur at $R = 0.9091$, which lies 0.009 above the PIGD threshold of 0.90;
- **26.2%** occur at $R = 1.1364$, which lies 0.014 below the TD threshold of 1.15;
- together these two values account for **78.2%** of the Indeterminate visits with a computable ratio.

The remaining nine values are spread thinly through the middle of the range, none of them accounting for more than 11% of observations. Figure 1 shows this distribution in its left panel, with each bar split by the subtype recorded at the patient's previous visit. Its right panel carries the accompanying next-visit breakdown from the same analysis, which we are not discussing here.

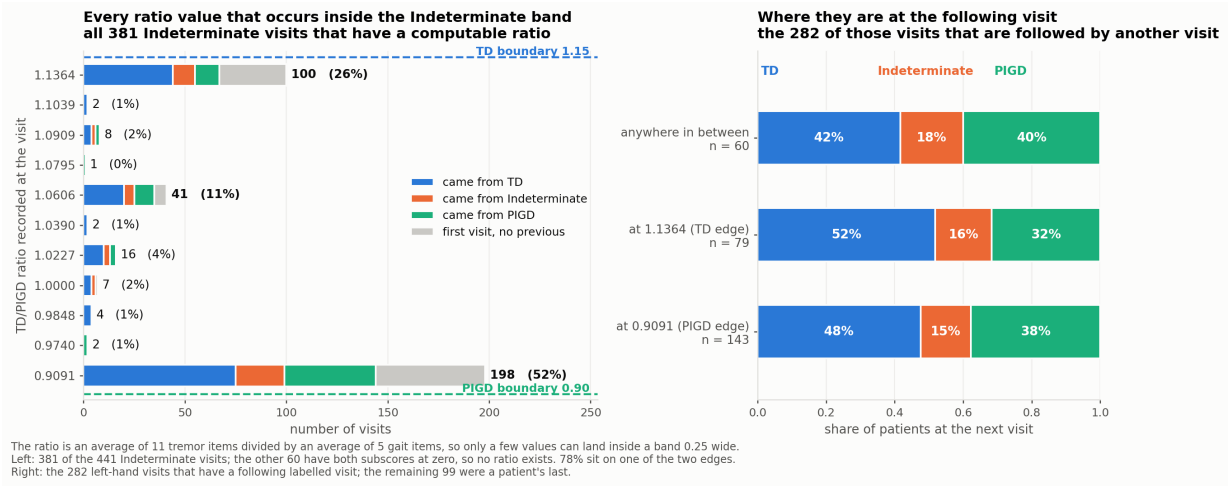


Figure 1: Distribution of the tremor/PIGD ratio among visits recorded as Indeterminate. *Left:* every ratio value that occurs inside the range $0.90 < R < 1.15$, across the 381 Indeterminate visits for which the ratio can be computed. Bar length gives the number of visits and colour shows the subtype recorded at the patient's previous visit, with visits that have no preceding labelled visit shown separately in grey. The dashed lines mark the PIGD threshold at 0.90 and the TD threshold at 1.15. *Right:* for the 282 of those visits that are followed by another labelled visit, the subtype recorded at that next visit, grouped by where in the range the patient's ratio was sitting. The figure was produced in response to the suggestion to examine where inside the Indeterminate range these observations fall. The left panel shows a strong concentration at the two values closest to the lower and upper boundaries; the right panel is included as part of the analysis and is left for discussion.

Broken down by where the patient came from, of the 287 Indeterminate visits that follow an earlier labelled visit, 165 follow a visit recorded as TD, 75 follow a visit recorded as PIGD and 47 follow another Indeterminate visit. The concentration at the two boundary values is present in all three groups and in the visits with no preceding label.

We are reporting this distribution as an observation and leaving its interpretation open for discussion.

3 Patient-level subtype progression

Our earlier figures showed the share of the assessed population in each subtype at each follow-up year. That summary has a known weakness: patients enter and leave the study at different times, so a shift in the proportions mixes real change with changes in who is being assessed. The request was to show individual patients instead.

A correction to the population used

Producing the patient-level version made us re-examine how we had been counting subtype changes, and we found an error in the earlier figure.

There are 1,080 patients with at least one labelled visit, but 146 of them have exactly one. A patient seen once cannot show a subtype change, because there is no second observation to compare against. Our earlier calculation nevertheless included them, and because their record collapses to a single subtype they were counted among the patients who never changed. That inflated the apparent stability of the cohort.

The correct population for any question about change is the **934 patients with at least two labelled visits**. All patient-level results below use that group, and the earlier figure has been regenerated.

Recorded subtype changes	Patients	Share
None	476	51.0%
One	210	22.5%
Two or more	248	26.6%
Total	934	100.0%

Table 1: Frequency of recorded subtype change among the 934 patients with at least two labelled visits. The 146 patients with a single labelled visit are excluded because they have no opportunity to show a change.

It is worth being explicit about what the first row means. "No recorded change" means that the same subtype label was assigned at every labelled visit the patient had. It does not mean the patient's condition was unchanged. A patient can worsen substantially and still stay on the same side of the threshold, and patients with short follow-up have fewer opportunities to be seen changing. The figure is best read as a statement about the record rather than about the disease.

Individual trajectories

Figure 2 shows every one of the 934 patients as a single row. This is the direct answer to the request: nothing is averaged, and no patient is left out.

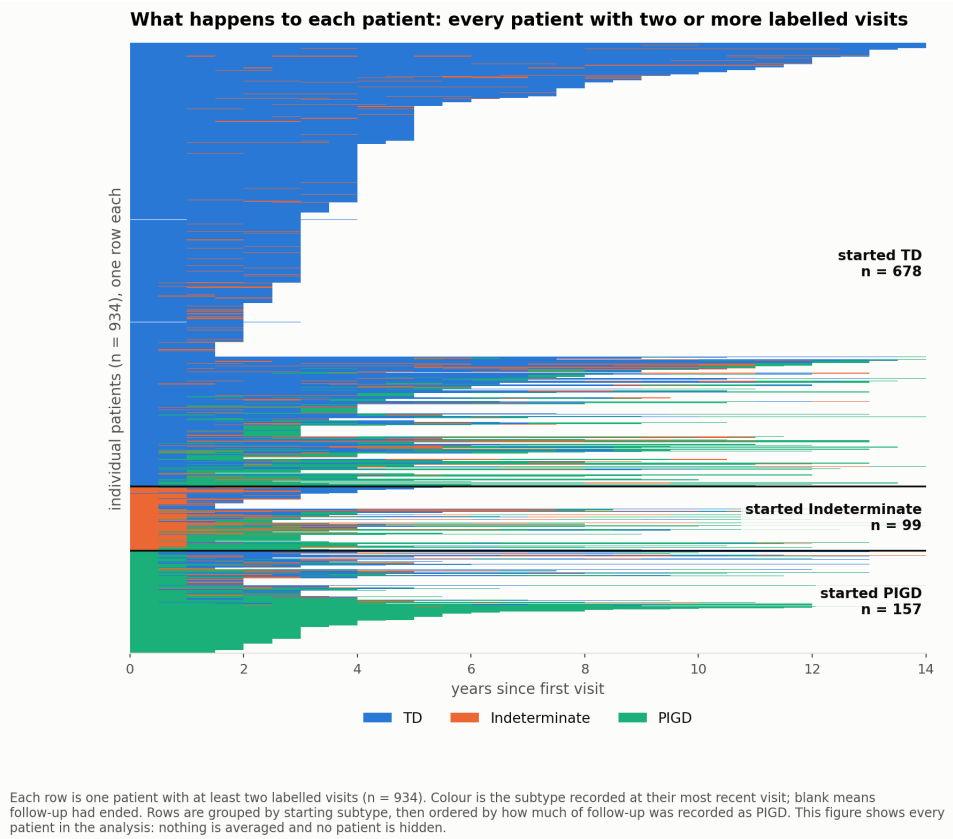


Figure 2: Motor subtype trajectories for all 934 patients with at least two labelled visits. Each horizontal row is one patient; colour shows the subtype recorded at that patient's most recent visit, and blank space means follow-up had ended. Rows are grouped by the subtype recorded at the first visit and then ordered by how much of the follow-up period was recorded as PIGD. Reading down each block shows how varied individual courses are, and the staircase shape on the right is the loss of patients to follow-up, which is what made the earlier population-level summary difficult to interpret.

Recorded subtype courses

A complementary way to summarise the same data is by the *course* a patient follows: the sequence of subtype labels across their visits, with repeats collapsed, so that a patient recorded as TD, TD, PIGD, PIGD has the course TD → PIGD. There are 86 distinct courses among the 934 patients. The twelve commonest cover 775 patients (83.0%); the remaining 159 patients are spread across 74 rarer courses.

Recorded course	Patients	Share
TD	399	42.7%
TD → PIGD	83	8.9%
PIGD	71	7.6%
TD → Indeterminate	39	4.2%
TD → Indeterminate → TD	35	3.7%
Indeterminate → PIGD	27	2.9%
Indeterminate → TD	26	2.8%
PIGD → TD	24	2.6%
TD → Indeterminate → PIGD	21	2.2%
TD → PIGD → TD	20	2.1%
PIGD → TD → PIGD	19	2.0%
Indeterminate → TD → PIGD	11	1.2%
Twelve commonest courses	775	83.0%

Table 2: The twelve commonest recorded subtype courses among the 934 patients with at least two labelled visits. Courses ending where they started, such as TD → Indeterminate → TD and PIGD → TD → PIGD, together account for a noticeable share of patients.

Figure 3 draws one representative patient for each of these twelve courses, so that what a course looks like in time can be seen directly. The representatives were picked as the patient with the typical number of visits for that course rather than the longest followed, because choosing the longest would make a decade of follow-up look normal when the median patient contributes three visits over about two years.

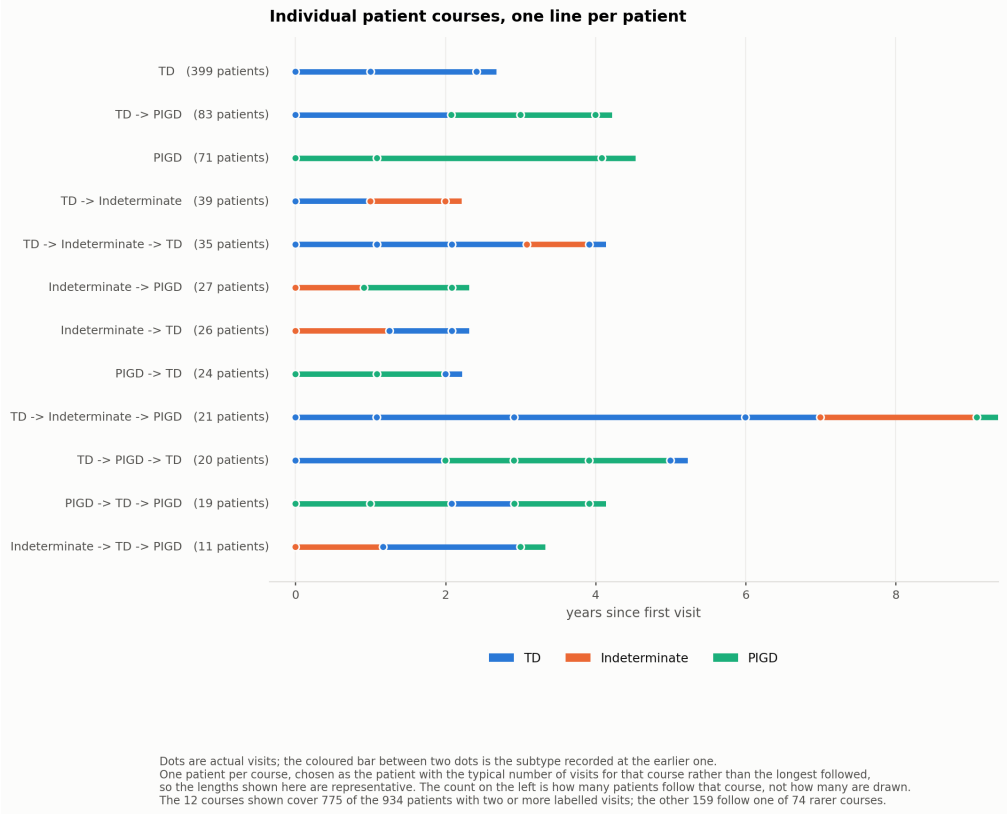


Figure 3: One representative patient for each of the twelve commonest recorded subtype courses. Dots mark actual visits and the coloured bar between two dots shows the subtype recorded at the earlier of them. The number beside each row is how many of the 934 patients follow that course, not how many patients are drawn; one patient is drawn per course. Note the short follow-up in most rows, which is typical of the cohort.

Two things stand out in Table 2. Nearly half the patients have a course consisting of a single subtype, which is consistent with Table 1. And among those who do change, courses that return to where they started are common: TD → Indeterminate → TD, PIGD → TD → PIGD and TD → PIGD → TD together account for 74 patients. This is the same instability we had already reported, seen from the level of the individual patient.

4 Conversion towards PIGD and towards TD

Our earlier analysis estimated how long it takes for a patient to be recorded as PIGD. The request was to run the same analysis in the opposite direction. We did this using exactly the same method, so that the two are comparable: patients count as being at risk if their first labelled visit is not the target subtype, conversion is their first visit recorded as that subtype, patients who never convert are censored at their last labelled visit, and a patient who moves back and forth is counted only at their first qualifying entry.

Two definitions of conversion are reported in both directions. Under the *single-visit* definition, one visit recorded as the target subtype counts. Under the *sustained* definition, the target subtype must still be recorded at the following visit. The gap between them measures how much of the apparent conversion consists of one-off assessments that are not repeated.

Direction	Definition	At risk	Conversions	5-year estimate	Median time
towards PIGD	single visit	885	263	35.3%	3.00 years
towards PIGD	sustained	885	127	15.3%	3.00 years
towards TD	single visit	304	125	61.2%	1.09 years
towards TD	sustained	304	60	33.9%	1.33 years

Table 3: Recorded conversion in both directions, estimated by the same survival method. The two at-risk groups are different populations and different sizes, so the percentages are not a head-to-head comparison.

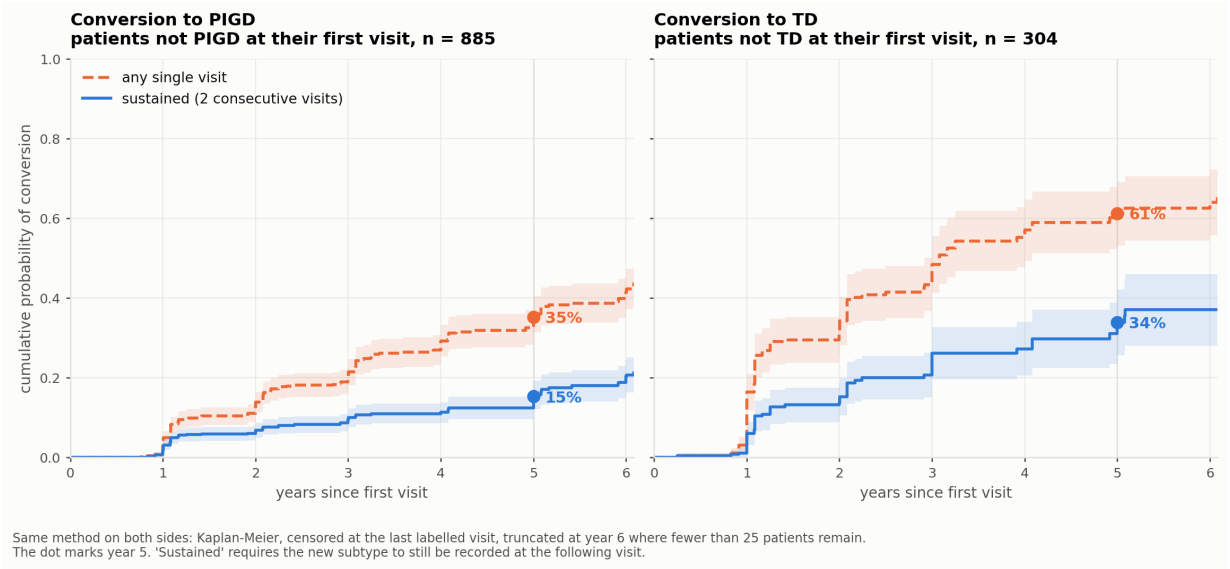


Figure 4: Estimated cumulative probability of being recorded in the target subtype over follow-up, for conversion towards PIGD (left) and towards TD (right), under both definitions. Both panels use the same method, the same axes and the same truncation point, so the shapes can be compared even though the two at-risk groups differ. The dot marks the five-year estimate. Shaded bands are 95% intervals. The right-hand panel was added in response to the request to examine conversion in the reverse direction.

Three points about Table 3 need stating carefully.

First, the two rows for each direction differ by a factor of roughly two. Requiring a subtype to still be recorded at the next visit more than halves the estimated five-year rate in both directions. Much of what looks like conversion under the looser definition is a single assessment that is not repeated.

Second, the at-risk groups are not the same. The group at risk of being recorded as PIGD is 885 patients; the group at risk of being recorded as TD is 304, because most of the cohort starts as TD. The 61.2% and 35.3% figures are therefore not directly comparable, and we would not want them presented side by side as though they were.

Third, and most important, a recorded conversion towards TD should not be read as clinical improvement. Under the single-visit definition, 61.2% of the relevant at-risk group were recorded as TD within five years, and half of those within about one year. But the group at risk consists of patients whose first visit was recorded as PIGD or Indeterminate, and the Indeterminate label in particular is the least reliable of the three. A substantial part of this movement is likely to be a first assessment that was not representative, corrected at the second visit, rather than a patient whose symptoms genuinely shifted. The observed subtype can also change again later, so a single TD observation is not a permanent state.

5 The raw tremor/PIGD ratio over follow-up

The subtype label is produced by cutting a continuous measurement into three categories, so any statement about label stability is really a statement about that measurement. Our earlier figure of ratio progression showed a fitted model, and the request was to see the measurements themselves. Figure 5 does that: every observation is plotted, each patient's visits are joined, and no model is involved.

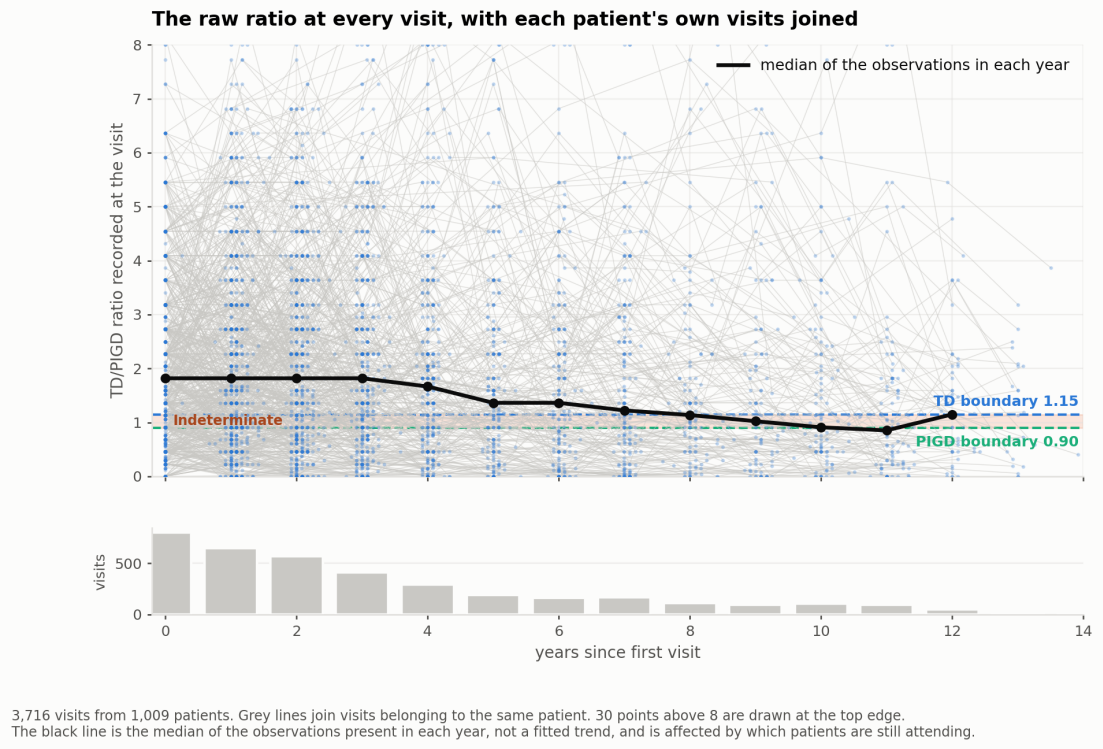


Figure 5: Raw observed tremor/PIGD ratio at every visit where it can be computed: 3,716 visits from 1,009 patients. Grey lines join visits belonging to the same patient. The dashed lines are the TD threshold at 1.15 and the PIGD threshold at 0.90, with the Indeterminate range shaded between them. The black line is the median of the observations actually present in each year; it is a summary of the raw data, not a fitted trend, and is affected by which patients are still attending. The lower panel gives the number of visits contributing to each year. No model output appears anywhere in this figure.

Two things can be read from it. There is a downward trend: the median observed ratio sits at about 1.82 through the first three years and falls to about 0.91 by year 10. That is consistent with the direction we had estimated previously, and it is visible in the measurements without any modelling. But the later part of that line rests on very few patients — by year 12 only 52 visits remain — and the median rises again there, which is a change in who is still attending rather than a change in the patients. The count panel is part of the reading.

The second and more important thing is how much scatter there is. The individual patient lines cross the Indeterminate band repeatedly and in both directions. To quantify this:

- the standard deviation of the ratio *within* a patient over time is about 1.22, against an Indeterminate range only 0.25 wide;
- about 54% of the total variation in the ratio is between patients and 46% is within a patient over time, an intraclass correlation of 0.536;
- the median change in a patient's own ratio from one visit to the next is 0.68, and in 73% of consecutive visit pairs the ratio moves by more than the entire width of the Indeterminate range.

Figure 6 shows the last of these directly.

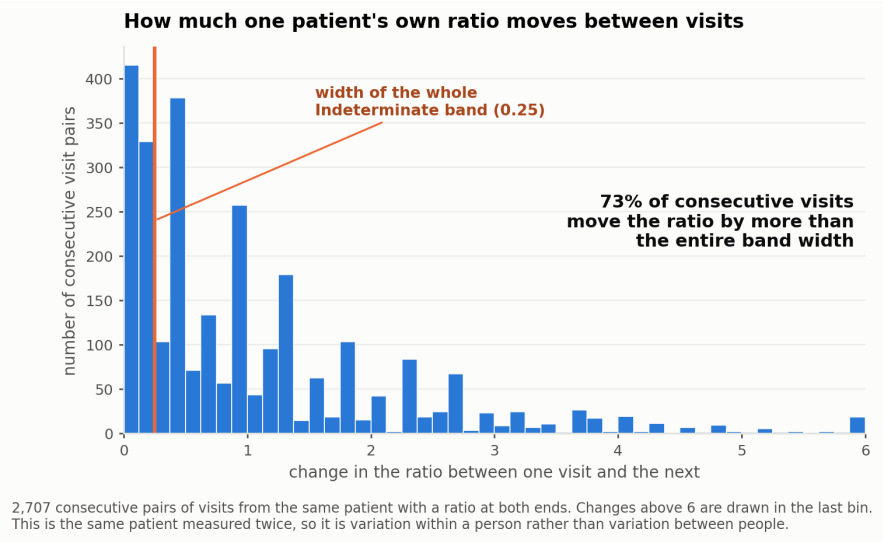


Figure 6: Size of the change in a single patient's own tremor/PIGD ratio between one visit and the next, across 2,707 consecutive visit pairs. The vertical line marks the width of the whole Indeterminate range for comparison. Because this compares the same patient measured twice, it isolates variation within a person from variation between people.

The practical consequence is the one that matters for how the subtype labels should be read. If the ratio routinely moves by more than the width of the band between two visits of the same patient, then a patient whose ratio happens to sit near a threshold will cross it from time to time for reasons unrelated to progression. A single recorded change of subtype is therefore weak evidence on its own, which is also what the gap between the two conversion definitions in Table 3 suggests.

6 Combined population-level view of the model results

The last request was a presentation one. The model-based population results had been drawn as three separate panels, one for each starting subtype, which makes it hard to compare them. Figure 7 puts all three on one set of axes. This is the same model and the same numbers as before, drawn differently; nothing about the model itself has changed.

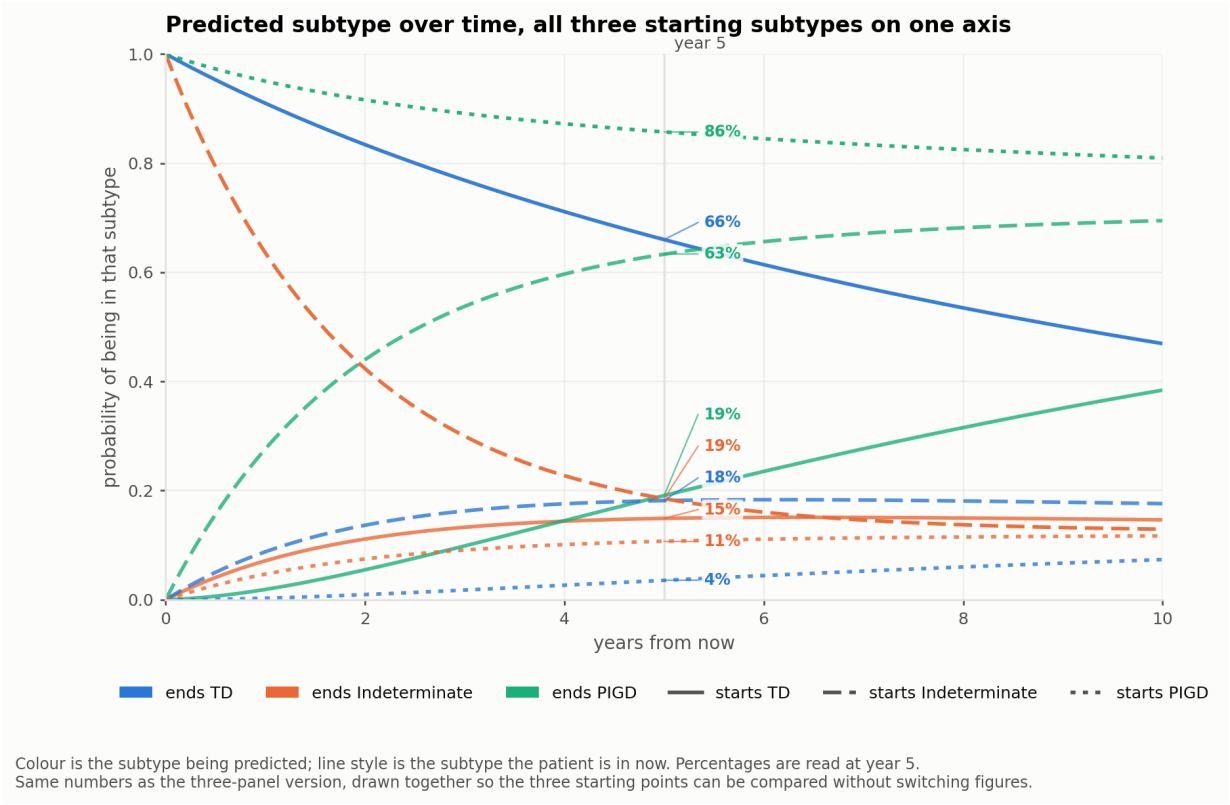


Figure 7: Model-based probability of occupying each subtype over time, for patients starting in each of the three subtypes, drawn on one set of axes. Colour shows the subtype being predicted and line style shows the subtype the patient starts in. Percentages are read at year 5. These are probabilities produced by the fitted model for a patient's underlying subtype, not observed proportions in the cohort. Replacing the earlier three-panel version allows the three starting subtypes to be compared without moving between figures.

Read at five years, the model gives:

Starting subtype	TD at 5 years	Indeterminate at 5 years	PIGD at 5 years
TD	66%	15%	19%
Indeterminate	18%	19%	63%
PIGD	4%	11%	86%

Table 4: Model-based probabilities at a five-year horizon, by starting subtype.

Putting the three on one axis makes the asymmetry easy to see. A patient in TD now has roughly a one-in-five chance of being in PIGD in five years, while a patient in PIGD now has about a one-in-twenty-five chance of being in TD. Patients currently in Indeterminate resemble PIGD far more than TD, with 63% of them in PIGD at five years. This is the same directional drift we see in the observed transitions, expressed as a probability over a fixed horizon.

Two caveats attach to this figure. The probabilities refer to the model's estimate of a patient's underlying subtype, not to the label that would be recorded at a visit, and they are not observed proportions in the cohort. And the projection assumes the fitted transition behaviour continues to apply over the whole horizon; we have some evidence that rates are not constant over follow-up, so the five-year values should be treated as indicative rather than exact.

7 What these analyses taken together suggest

The five pieces of work were requested separately, but they point in a consistent direction.

About half of the patients who can be assessed show no recorded subtype change at all during their follow-up, and the other half change at least once, often more than once and often returning to where they started. Separately, the Indeterminate observations were found to cluster at the two values closest to the boundaries of the Indeterminate range. That distribution is recorded here as an observation; what should be made of it is left for discussion.

The conversion analysis run in both directions shows that movement is recorded towards TD as well as towards PIGD, and that requiring a change to persist to the following visit roughly halves the estimated rate in both directions. The raw ratio explains why: within a single patient the measurement moves by more than the width of the Indeterminate range between most pairs of consecutive visits.

The practical conclusion is about interpretation rather than about the disease. A single recorded change of subtype is not strong evidence that a patient's condition has changed, and a change that persists across two visits should be given considerably more weight than one that does not. Where a result depends on counting subtype changes, it is worth checking how sensitive it is to that distinction. This applies to our own earlier results as much as to anything else, and Table 3 is the clearest illustration of the size of the effect.

Separately, correcting the population used for the patient-level analysis matters in its own right. Restricting to patients who can actually demonstrate a change lowered the proportion recorded as never changing, and we have used the corrected figure of 51.0% throughout.

8 Current status and the next validation step

All five of the requested analyses are complete. The distribution of ratios within the Indeterminate range was examined and is reported in Section 2 for discussion. The patient-level progression analysis was corrected and regenerated on the appropriate population of 934 patients. Conversion towards TD was added using the same method as the existing PIGD analysis. The raw ratio was examined directly, without a model. The population-level model results were combined into a single figure.

A stricter patient-level leave-one-out validation of the existing hidden Markov model is currently being performed as a separate analysis. In that analysis each patient is held out completely, the model is fitted using the remaining patients, and the held-out patient's next recorded subtype is predicted one visit at a time using only the history available before each prediction. Those results will be reported separately.

Data used in the preparation of this report were obtained from the Parkinson's Progression Markers Initiative (PPMI) database.